

Wassim Ait Oubahou

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SUMMARY

MEng Mechanical Engineer specialising in structural simulation and mechanical design. Delivered validated FEA and CAD packages for industrial clients; currently leading the structural, thermal, and orbital analysis of an independent 16U CubeSat.

EDUCATION

University of Derby

Jan 2018 – Jun 2022

MEng Mechanical Engineering (2:1, with Honours)

Derby, UK

- **Master's Thesis:** Mechanical and vibrational characterisation of CFRP/flux hybrid composites. Improved impact resistance by 22% and structural damping by 17% through optimised fibre stacking sequences.
- **Third-Year Project:** Computational modelling of hydrodynamics in porous media.

WORK EXPERIENCE

Freelance Mechanical Design Engineer

Jan 2023 – Present

Self-employed

Agadir, Morocco

- Delivered a complete ski binding redesign for an equipment manufacturer: full GD&T drawing set, STEP export, and FEA package (static, modal, thermal). Reduced component mass by 22% while maintaining SF > 2.5 under worst-case loading.

Mechanical Engineer – Intern

Jun 2018 – Aug 2018

COREteQ Systems

Frimley, UK

- Contributed to the redesign of an ESP component using SolidWorks and ANSYS, extending service life and achieving compliance with relevant industry standards.

ENGINEERING PROJECTS

Spacecraft Systems Engineer – Independent Project

Sep 2022 – Present

AtlasSAT – 16U CubeSat for Earth Observation (documentation available on GitHub)

Agadir, Morocco

- **Mission & CAD:** 16U Earth Observation CubeSat (~23 kg, SSO 550 km) for water resource monitoring. 8-subsystem architecture; fully integrated SolidWorks assembly with GD&T drawings and $\text{\LaTeX}$ technical reports.
- **Structural FEA:** Quasi-static and modal analyses per Falcon 9 Rideshare RPUG r2.0 & ECSS-E-ST-32C. Primary chassis FS = 118; first modal freq. $f_1 = 503$ Hz (req. >140 Hz); 400-mode extraction achieving >90% effective mass participation.
- **Thermal analysis:** Steady-state lumped-parameter model and Python transient simulation for hot/cold orbital cases. Cold case $\Delta T = 17$ K; hot case $\Delta T = 96$ K across the primary structure.
- **Orbital & power:** GMAT SSO ground-track and eclipse simulation; MATLAB/Simulink power budget model for subsystem allocation and battery sizing.

Mechanical Design Engineer – Academic Project

Feb 2018 – Jun 2022

BMFA Payload Challenge

Derby, UK

- Designed, fabricated, and flight-tested a fixed-wing UAV achieving 6+ minutes of flight with full payload; SolidWorks design, laser-cut and 3D-printed structure.
- Validated aerodynamic stability through wind tunnel testing and field trials; airframe met all competition performance requirements.

AWARDS

Peter Parkin Memorial Lecture 2021 – Nominee: Presented research on hybrid reinforced composites for railway structural applications.

TECHNICAL SKILLS

CAD / Design: SolidWorks (assemblies, GD&T drawings), AutoCAD

FEA / CFD: SolidWorks Simulation, ANSYS (Static Structural, Modal, Thermal), Star-CCM+

Programming: Python (numerical simulation, scripting), MATLAB, Simulink, C++, Git

Mission & Systems: GMAT (orbital mechanics)

Machining: lathe, milling machine, laser cutting, 3D printing

Languages: French (fluent), English (fluent), Moroccan Arabic (native), Mandarin (basic)